$$\angle A + \angle QNR = 180^{o} \qquad (6.85)$$

In the triangle ΔQNR,

$$r_1 + r_2 + \angle QNR = 180^{o} \qquad (6.86)$$

Comparing the two equations (6.85) and (6.86) we get,

$$r_1 + r_2 = A \qquad (6.87)$$

Substituting this in equation (6.84) for angle of deviation,

$$d = i_1 + i_2 - A \qquad (6.88)$$

Thus, the angle of deviation depends on the angle of incidence i_1, angle of emergence i_2 and the angle for the prism A. But, for a given angle of incidence, the angle of emergence is decided by the refractive index of the material of the prism. Hence, the angle of deviation for a monochromatic light depends on these following factors.

(i) the angle of incidence

(ii) the angle of the prism

(iii) the refractive index of the material of the prism (which decides the angle of emergency).

EXAMPLE 6.19

A monochromatic light is incident on an equilateral prism at an angle 30° and is emergent at an angle of 75°. What is the angle of deviation produced by the prism?

Solution

Since, the prism is equilateral, $A = 60°$;

Given, $i_1 = 30°$; $i_2 = 75°$

Equation for angle of deviation, $d = i_1 + i_2 - A$

Substituting the values, $d = 30°+75°-60°=45°$

The angle of deviation produced, $d = 45°$

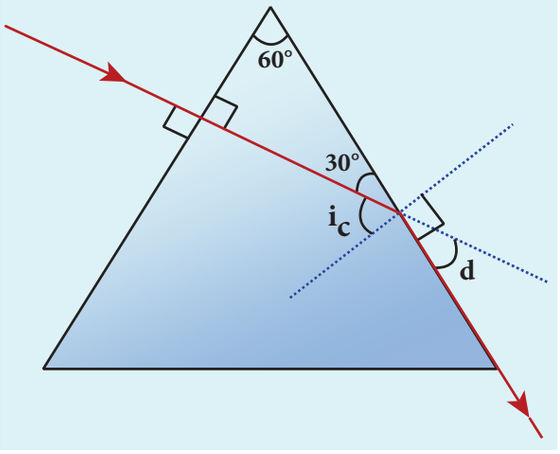
EXAMPLE 6.20

Light ray falls at normal incidence on the first face and emerges gracing the second face for an equilateral prism.

(a) What is the angle of deviation produced?

(b) What is the refractive index of the material of the prism?

Solution



The given situation is shown in the figure.

Given, $A = 60°$; $i_1 = 0°$; $i_2 = 90°$

(a) Equation for angle of deviation,

$d = i_1 + i_2 - A$

Substituting the values,

$d = 0° + 90° - 60° = 30°$

The angle of deviation produced is, $d = 30°$

(b) The light inside the prism must be falling on the second face at critical angle as it graces the boundary. $i_c = 90° - 30° = 60°$

Equation for critical angle is, $\sin i_c = \frac{1}{n}$

$$n = \frac{1}{\sin i_c}; \quad n = \frac{1}{\sin 60^{\circ}} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}} = 1.15$$

The refractive index of the material of the prism is, $n = 1.15$

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